

DA 515. Spring 2026

Machine Learning Final Project Proposal

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Project Title: Synthetic Speech Detection & Attribute Analysis

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A. Project Goal

Develop a robust binary classification system to distinguish authentic human speech from AI-generated (synthetic) audio. Beyond simple classification accuracy, this project evaluates how different architectural approaches (traditional ML vs. deep learning) handle spectral artifacts commonly found in deepfake audio. This comparison will illuminate which models best capture the subtle acoustic signatures that distinguish genuine from manipulated speech, improving detection reliability in real-world scenarios.

B. Dataset Description

Dataset: ASVspoof 2019 (<https://huggingface.co/datasets/asvspoof/asvspoof2019>).

This gold-standard dataset contains thousands of audio samples: bona fide (genuine human) speech and spoofed audio generated using state-of-the-art Text-to-Speech (TTS) and Voice Conversion (VC) technologies. Raw PCM audio files will be preprocessed into LFCC (Linear Frequency Cepstral Coefficients) and MFCC (Mel-Frequency Cepstral Coefficients) to capture physiological and mathematical nuances of audio signals.

C. ML Algorithms & Tools

Three complementary ML approaches will be implemented and compared:

- 1. LightGBM:** High-performance gradient boosting on tabular MFCC/LFCC features. Serves as a baseline for traditional supervised learning.
- 2. Convolutional Neural Networks (ResNet/LCNN):** Treats spectrograms as 2D images to learn spatial-temporal patterns indicative of synthetic manipulation.
- 3. Gaussian Mixture Models (GMM):** Probabilistic baseline for modeling acoustic feature distributions, commonly used in biometric verification.

Evaluation Metrics: Equal Error Rate (EER), Accuracy, and F1-score to comprehensively assess model performance.

Tools: Librosa/Torchaudio (feature extraction), Scikit-learn (preprocessing and GMM), PyTorch (CNN design and training).

This project leverages a curated, real-world dataset with three diverse ML paradigms to advance synthetic speech detection and understand the trade-offs between interpretability and performance in audio forensics.